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B.Tech. Degree VIII Semester Supplementary Examination July 2019

CS 15-1802 ARTIFICIAL INTELLIGENCE (2015 Scheme)

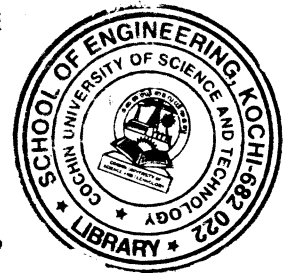
Time: 3 Hours

Maximum Marks: 60

PART A (Answer ALL questions)

(10 × 2 = 20)

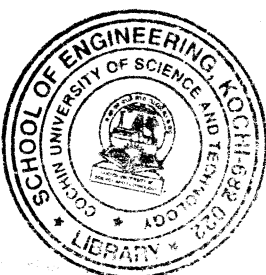
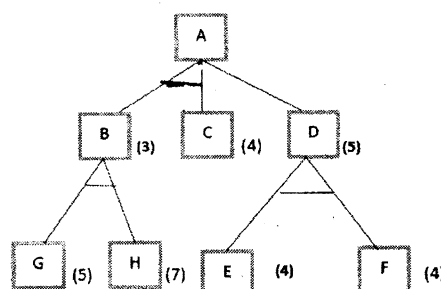
- I. (a) Explain the working of a software agent.
- (b) Describe agent environment properties. Write the environment properties of chess game.
- (c) Draw the first three levels of BFS and DFS tree for the water jug problem.
- (d) What are the disadvantages of hill climbing? Write methods to overcome the problems.
- (e) What is a clause? Write the important steps in conversion to clause form.
- (f) Find the most general unifier for the following pairs of predicates
 - (i) $g(f(x), y, joe) \quad g(z, w, v)$
 - (ii) $g(f(x)) \quad g(x)$
- (g) How is heuristics useful in a search process? How heuristics can be added to a search procedure? Give one sample heuristic function.
- (h) What are the building blocks of a production system model.
- (i) Construct semantic net representation for the following
 - (i) Pompeian(Marcus)
 - (ii) Blacksmith(Marcus)
- (j) Can we use Bayesian networks as a basis for probabilistic reasoning? Justify your answer with an example.



PART B

(4 × 10 = 40)

- II. (a) Explain the working of a rational agent. (3)
 - (b) Analyse the working of a goal based software agent with a utility based software agent. (7)
- OR**
- III. (a) What is agent oriented programming. (3)
 - (b) Describe java implementation of intelligent agents. (7)
- IV. (a) Explain AO* algorithm. (5)
 - (b) Show the different steps of application of AO* algorithm for the AND-OR graph given below. (5)



(P.T.O.)

OR

- V. (a) Explain hill climbing procedure. (5)
 (b) Explain hill climbing procedure in the case of blocks world problem. (5)

- VI. (a) Represent the following sentences in first order predicate logic. (5)
 (i) Lucy is a professor
 (ii) All professors are people.
 (iii) Joe is a dean
 (iv) Deans are professors
 (v) All professors consider dean as a friend or don't know him
 (vi) people criticize people who are not friends
 (vii) Lucy criticised Joe
 (b) Prove that Lucy is not a friend of Joe using backward chaining. (5)

OR

- VII. (a) Explain the basics of resolution procedure (2)
 (b) Describe the resolution algorithm (4)
 (c) $P \rightarrow Q$ (4)
 1. $(Q \wedge S) \rightarrow R$
 2. $\neg R$

Prove that P is false using resolution

- VIII. (a) Represent the following sentence in case grammars. (2)
 (i) John flew to New York
 (ii) The co-pilot flew the plane
 (b) Explain in detail syntactic analysis step and semantic analysis step of natural language processing. (8)

OR

- IX. (a) Describe a membership function in fuzzy logic with a suitable example. (3)
 (b) What is learning? Describe various methods of learning. (7)
